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1) Error is important in the computational science because many iterative method is based on error based calculation. The methods such as Newton Raphson, Bisection, Gauss-seidel use the error based calculation. Error also helps to maximize the accuracy of the accuracy of the problem.

Fields such as physics, computer science, economics etc. use the study of errors. Error in physics is also used in finding accuracy in the given solution. For example, Vernier caliper is used to determine the radius of cylinder using the error. This experiment shows that nobody is perfect in calculation. Error is used in computer science to ~~redu~~ increase the efficiency of algorithm. As I mentioned earlier. Newton Raphson, Bisection are used to minimize the error. Newton Raphson uses the error minimization in solving the problem.

Absolute error is the calculation of the final value with the initial value divided by final value multiply by 100%.

$$|E_x| = \left| \frac{\text{Final value} - \text{Initial value}}{\text{Final value}} \right| \times 100\%$$

Let us take the final value be 1.356 using the Newton Raphson method and the initial value be 1

Now
Using the formula,

$$|E_x| = \left|\frac{\text{final value} - \text{Initial value}}{\text{final value}}\right| \times 100\%$$

$$= \left|\frac{1.356 - 1}{1.356}\right| \times 100\%$$

$$= |0.2625| \times 100\%$$

$$= 26.25\%$$

This was found out that the error percentage is 26.25% in the Newton Raphson method.

$$\frac{d(\cos x)}{dx} + \frac{de^x}{dx} - \frac{d(4)}{dx}$$

) Let $f(x) = \cos x + e^x - 4 = 0$ $= e^x - \sin x$

Now,

$f'(x) = -\sin x + e^x$

Let $x_0 = 0$

Iteration I Now,

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 0 - \frac{\cos 0 + e^0 - 4}{-\sin 0 + e^0}$$

$$x_1 = 0 - \frac{1+1-4}{-0+1}$$

$$x_1 = 0 - \frac{2-4}{1}$$

$$x_1 = 0 - (-2)$$

$$x_1 = 2$$

Now,

$$|E_a|_x = \left|\frac{2-0}{2}\right| \times 100\%$$

$= 100\%$

Iteration 2

$x_1 = 2$

Now,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 2 - \frac{\cos 2 + e^2 - 4}{-\sin 2 + e^2}$$

$$x_2 = 1.5412$$

Now,

$$|\epsilon_n| = \left| \frac{1.5412 - 2}{1.5412} \right| \times 100\%$$

$$= 29.76\%$$

Iteration 3

$$x_2 = 1.5412$$

Now

$$x_3 = 1.5412 - \frac{\cos(1.5412) + e^{1.5412} - 4}{-\sin 1.5412 + e^{1.5412}}$$

$$x_3 = 1.35055$$

Now

$$|\epsilon_n| = \left| \frac{1.35055 - 1.5412}{1.35055} \right| \times 100\%$$

$$= 14.11\%$$

Iteration 4

$$x_3 = 1.35055$$

Now

$$x_4 = 1.35055 - \frac{\cos(1.35055) + e^{1.35055} - 4}{-\sin(1.35055) + e^{1.35055}}$$

$$x_4 = 1.3234$$

$$|\epsilon_a| = \left| \frac{1.3234 - 1.35055}{1.3234} \right| \times 100\%$$

$$= 2.05\%$$

$\therefore$ The root is 1.3234 after 4 iteration

4) Soln:-

	x_0	x_1	x_2	x_3
x	1.0	1.1	1.2	1.3
$(\cos x+1)$	1.5403	1.4536	1.3624	1.2675
	y_0	y_1	y_2	y_3

Now,
The function is $\cos(1.15)$
$f(x) = \cos(1.15)$
$f(x) = \cos(x)$
Comparing we get,
$x = 1.15$

Now,
Using Lagrange Interpolation we get,

$$L(x) = \frac{(1.15-1.1)(1.15-1.2)(1.15-1.3)}{(1.0-1.1)(1.0-1.2)(1.0-1.3)} \times 1.5403 + \cancel{(1.15-1.0)(1.15-1.2)(1.15-1.3)}$$

$$\frac{(1.15-1)(1.15-1.2)(1.15-1.3)}{(1.1-1.0)(1.0-1.2)(1.0-1.3)} \times 1.4530 +$$

$$\frac{(1.15-1)(1.15-1.1)(1.15-1.3)}{(1.2-1)(1.2-1.1)(1.2-1.3)} \times 1.3624 +$$

$$\frac{(1.15-1)(1.15-1.1)(1.15-1.2)}{(1.3-1)(1.3-1.1)(1.3-1.2)} \times 1.2675$$

$$= \frac{-1}{16} \times 1.5403 + \frac{3}{16} \times 1.4530 + \frac{9}{16} \times 1.3624 +$$

$$1.2675 \times \left(-\frac{1}{16}\right)$$

$$= 0.8627$$

Now,

3) Soln:

The given equation are:-

$$x + 2y - z = 3 \quad \text{---(i)}$$
$$2x + 4y + z = 12 \quad \text{---(ii)}$$
$$x + y + 2z = 9 \quad \text{---(iii)}$$

Arranging the equation in matrix form we get,

$$\left[\begin{array}{ccc:c} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 12 \\ 1 & 1 & 2 & 9 \end{array}\right]$$

$\cancel{R_2 \to R_2 - 2R_1}$

$\cancel{R_3 \to R_3 - R_1}$

$R_2 \leftrightarrow R_3$

$$\left[\begin{array}{ccc:c} 1 & 2 & -1 & 3 \\ 1 & 1 & 2 & 9 \\ 2 & 4 & 1 & 12 \end{array}\right]$$

$R_2 \to R_1 - R_2$

$R_3 \to R_3 - 2R_1$

$$\left[\begin{array}{ccc:c} 1 & 2 & -1 & 3 \\ 0 & 1 & -3 & -6 \\ 0 & 0 & 3 & 6 \end{array}\right]$$

$R_3 \to \frac{1}{3} R_3$

$$\left[\begin{array}{ccc|c} -1 & & & \\ 0 & 1 & -3 & -6 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & -3 & -6 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$R_2 \to R_2 + 3R_3$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$R_1 \to R_1 - 2R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$R_1 \to R_1 + R_3$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$\therefore$ Making matrix

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 2 \end{bmatrix}$$

$\therefore x = 5, \; y = 0, \; z = 2$